

Rule Engine – Logic, Flowchart & Decision Matrix

1. Overview

The Rule Engine decides:

- When irrigation should turn **ON**
- When it should turn **OFF**
- Which **pump mode** to use (Auto / Manual)
- How to use ET (Evapotranspiration) + soil moisture + weather forecast.

This document defines the logic clearly for developers, testers, and stakeholders.

2. High-Level Logic Flow

```
flowchart TD
    A[Sensor Data Received] --> B{Is Sensor Data Valid?}
    B -- No --> Z[Use last known valid values]
    B -- Yes --> C[Calculate ET using ET Module]
    C --> D[Check Soil Moisture Thresholds]
    D --> E{Soil Moisture < Lower Limit?}
    E -- Yes --> F[Check Weather Conditions]
    E -- No --> X[Turn OFF Pump]
    F --> G{Rain Expected Soon?}
    G -- Yes --> Y[Delay Irrigation by 2-6 hours]
    G -- No --> H[Turn ON Pump]
    H --> I{Reached Upper Moisture Limit?}
    I -- Yes --> X
    I -- No --> H
```

3. Decision Matrix

Condition	Moisture	ET	Rain Forecast	Action
Dry soil, no rain coming	< LowerLimit	High	No	Start irrigation
Dry soil but rain expected	< LowerLimit	Medium/High	Yes	Delay irrigation
Moisture in normal range	Between limits	Any	Any	Do nothing
High moisture	> UpperLimit	Low	Any	Stop irrigation

Condition	Moisture	ET	Rain Forecast	Action
Sensor error	N/A	N/A	N/A	Use last valid reading

4. Thresholds

Default recommended values:

Parameter	Value
Soil Moisture Lower Limit	30%
Soil Moisture Upper Limit	70%
Temperature Range Validity	0–55°C
ET High Level	> 4.0 mm/day
ET Medium	2.0–4.0 mm/day
ET Low	< 2.0 mm/day

5. Rule Engine Pseudocode

```
def rule_engine(data, last_state):
    moisture = data["soil_moisture"]
    et = data["et_value"]
    rain = data["rain_forecast"]
    pump_state = last_state["pump"]

    # 1. Validate incoming data
    if not data["valid"]:
        return last_state # keep old state

    # 2. If soil too dry
    if moisture < LOWER_LIMIT:
        if rain:
            return {"pump": "DELAY", "reason": "Rain expected"}
        return {"pump": "ON", "reason": "Soil dry"}

    # 3. If soil normal
    if LOWER_LIMIT <= moisture <= UPPER_LIMIT:
        return {"pump": "OFF", "reason": "Moisture OK"}

    # 4. If soil too wet
    if moisture > UPPER_LIMIT:
        return {"pump": "OFF", "reason": "Soil too wet"}

    return last_state
```

6. Historical Data Testing (Back-Testing)

You must verify rule engine performance using previous season data.

Steps:

- 1. Load CSV of historical moisture + ET + rain forecast.
- 2. Run rule engine across each day.
- 3. Log pump ON/OFF decisions.
- 4. Compare with farmer’s real irrigation patterns.
- 5. Adjust thresholds if system is too aggressive or too conservative.

Example back-test skeleton:

```
import pandas as pd

df = pd.read_csv("historical_data.csv")
state = {"pump": "OFF"}

for _, row in df.iterrows():
    decision = rule_engine(row, state)
    print(row["date"], decision)
    state = decision
```

7. Outputs the Rule Engine Must Produce

Field	Description
pump	ON / OFF / DELAY
reason	Why the decision was taken
timestamp	Server time
inputs_used	Sensors + ET values
safety_checks	Validation results

8. Error Handling

- Missing sensor data → fallback to cached values.
 - ET module failure → retry 3 times then use last ET value.
 - Weather API down → run without forecast.
 - Sensor out of bounds (example: moisture > 100%) → mark invalid.
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9. Future Improvements

- ML-based irrigation prediction model.

- Threshold auto-learning per crop type.
- Multi-zone irrigation rules.
- Fuzzy logic decision-making.